

Kārigar

AI-Driven Market Linkage & Smart Cataloguing for Marginalised Artisans
Smart India Hackathon · Problem Statement 26090

Research & Reference Bibliography

This document lists the literature that informed each feature (F1–F8) of the Kārigar prototype, organised by feature, with a one-line note on how each work was applied. It contains 52 references. Most are from 2018–2026 and appear in IEEE, ACM, and top-tier machine-learning venues (ICCV, CVPR, NeurIPS, ICML, KDD, SIGIR, CIKM, WWW, AAAI, ACL, IJCAI, CP, IEEE TPAMI / TIP / TKDE, IEEE/ACM TASLP). A small number of foundational method papers (the Shapley value, quantile regression, GrabCut, CLAHE) are cited at their primary source because that is the correct attribution for what the code actually does.

Note on use. This is a curated reading list compiled by the team to document design provenance. Bibliographic details (page numbers, volume / issue, exact venue year) should be verified against each publisher or DOI record before use in a formal submission.

Honesty scope. Where the prototype uses a deterministic or heuristic stand-in for a heavier model (GrabCut for SAM; a rule engine plus heuristic for CLIP-class visual entailment; a deterministic transcript when no ASR engine is installed), both the stand-in and the referenced production method are listed, and the distinction is stated in the app and its documentation.

A. F1 — AI Product Studio: segmentation, low-light enhancement, image-quality gating

Promptable segmentation to isolate the product, reference-free tone/colour correction, and no-reference quality scoring so a good artisan photo is never made worse.

[1] A. Kirillov, E. Mintun, N. Ravi, et al., “Segment Anything,” in Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV), 2023, pp. 4015–4026.

Applied in Kārigar for: Promptable product-vs-background segmentation (SAM); GrabCut is the prototype stand-in.

[2] C. Rother, V. Kolmogorov, and A. Blake, “GrabCut: Interactive Foreground Extraction Using Iterated Graph Cuts,” ACM Trans. Graphics (SIGGRAPH), vol. 23, no. 3, pp. 309–314, 2004.

Applied in Kārigar for: The iterated graph-cut segmentation actually used in the prototype.

[3] C. Guo, C. Li, J. Guo, et al., “Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2020, pp. 1780–1789.

Applied in Kārigar for: Zero-DCE-style tone-curve lift for under-exposed photos.

[4] K. Zuiderveld, “Contrast Limited Adaptive Histogram Equalization,” in Graphics Gems IV, Academic Press, 1994, pp. 474–485.

Applied in Kārigar for: The CLAHE local-contrast step, applied only when measured global contrast is low.

[5] C. Li, C. Guo, L.-H. Han, et al., “Low-Light Image and Video Enhancement Using Deep Learning: A Survey,” IEEE Trans. Pattern Analysis and Machine Intelligence, vol. 44, no. 12, pp. 9396–9416, 2022.

Applied in Kārigar for: Survey grounding the enhancement design choices and their failure modes.

[6] H. Talebi and P. Milanfar, “NIMA: Neural Image Assessment,” IEEE Trans. Image Processing, vol. 27, no. 8, pp. 3998–4011, 2018.

Applied in Kārigar for: Learned aesthetic/technical quality scoring — the model class behind the readiness gate.

[7] A. Mittal, A. K. Moorthy, and A. C. Bovik, “No-Reference Image Quality Assessment in the Spatial Domain,” IEEE Trans. Image Processing, vol. 21, no. 12, pp. 4695–4708, 2012.

Applied in Kārigar for: BRISQUE-style blind quality features (blur, noise) in the F1 component scores.

[8] J. Ke, Q. Wang, Y. Wang, P. Milanfar, and F. Yang, “MUSIQ: Multi-Scale Image Quality Transformer,” in Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV), 2021, pp. 5148–5157.

Applied in Kārigar for: Multi-scale no-reference IQA — documented upgrade path for the readiness score.

[9] G. Sharma, W. Wu, and E. N. Dalal, “The CIEDE2000 Color-Difference Formula: Implementation Notes, Supplementary Test Data, and Mathematical Observations,” Color Research & Application, vol. 30, no. 1, pp. 21–30, 2005.

Applied in Kārigar for: Colour-fidelity safety gate: reject an enhancement that shifts the product’s true colour.

B. F2 — Multilingual grounded auto-cataloguer: ASR, Indic NLP, cross-modal checking

Regional-language speech-to-text, structured attribute extraction, and evidence-based consistency checks so a listing never states a claim the artisan did not make.

[10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust Speech Recognition via Large-Scale Weak Supervision,” in Proc. Int. Conf. Machine Learning (ICML), 2023, pp. 28492–28518.

Applied in Kārigar for: Whisper — the real ASR engine wired into the prototype (faster-whisper / API path).

[11] T. Javed, S. Doddapaneni, A. Raman, et al., “Towards Building ASR Systems for the Next Billion Users,” in Proc. AAAI Conf. Artificial Intelligence, 2022, pp. 10813–10821.

Applied in Kārigar for: IndicWav2Vec — documented low-resource ASR upgrade covering 22 Indian languages.

[12] T. Javed, J. A. Nawale, E. I. George, et al., “IndicVoices: Towards Building an Inclusive Multilingual Speech Dataset for Indian Languages,” in Findings of the Association for Computational Linguistics (ACL), 2024.

Applied in Kārigar for: Benchmark and data design for evaluating regional-language voice input.

[13] J. Gala, P. A. Chitale, R. AK, et al., “IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages,” Trans. Machine Learning Research (TMLR), 2023.

Applied in Kārigar for: Regional-language ↔ English for the bilingual (EN/HI) catalogue output.

[14] S. Khanuja, D. Bansal, S. Mehtani, et al., “MuRIL: Multilingual Representations for Indian Languages,” arXiv:2103.10730, 2021.

Applied in Kārigar for: Encoder for KG-seeded slot / attribute extraction across Indic scripts.

[15] A. Radford, J. W. Kim, C. Hallacy, et al., “Learning Transferable Visual Models From Natural Language Supervision,” in Proc. Int. Conf. Machine Learning (ICML), 2021, pp. 8748–8763.

Applied in Kārigar for: CLIP-style image–text alignment; target model for the cross-modal authenticity check.

[16] N. Xie, F. Lai, D. Doran, and A. Kadav, “Visual Entailment: A Novel Task for Fine-Grained Image Understanding,” arXiv:1901.06706, 2019.

Applied in Kārigar for: Formal framing of the SUPPORTED / PARTIALLY_SUPPORTED / UNSUPPORTED verdict.

[17] Z. Ji, N. Lee, R. Frieske, et al., “Survey of Hallucination in Natural Language Generation,” ACM Computing Surveys, vol. 55, no. 12, pp. 1–38, 2023.

Applied in Kārigar for: Motivates the Fact-Grounding Guard — generate only from validated slots.

[18] R. Prabhavalkar, T. Hori, T. N. Sainath, R. Schlüter, and S. Watanabe, “End-to-End Speech Recognition: A Survey,” IEEE/ACM Trans. Audio, Speech, and Language Processing, vol. 32, pp. 325–351, 2024.

Applied in Kārigar for: Survey grounding the pluggable ASR architecture and its graceful fallbacks.

C. F3 — Craft knowledge graph and Digital Craft Passport (provenance)

A relational craft/technique/region graph with recursive traversal, and a provenance-confidence score that never conflates a sample record with a government certificate.

[19] A. Hogan, E. Blomqvist, M. Cochez, et al., “Knowledge Graphs,” ACM Computing Surveys, vol. 54, no. 4, pp. 1–37, 2021.

Applied in Kārigar for: Design of the craft / technique / material / region graph and its typed edges.

[20] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, “Unifying Large Language Models and Knowledge Graphs: A Roadmap,” IEEE Trans. Knowledge and Data Engineering, vol. 36, no. 7, pp. 3580–3599, 2024.

Applied in Kārigar for: KG + LLM roadmap for future passport reasoning and question answering.

[21] M. Feng, X. Wang, J. Zhang, et al., “Applying Blockchain Technology to Improve Agri-Food Traceability: A Review of Development Methods, Benefits and Challenges,” Journal of Cleaner Production, vol. 260, 121031, 2020.

Applied in Kārigar for: Provenance / traceability patterns for GI-linked craft goods, and their limits.

D. F4 — Fair, cost-aware pricing: gradient boosting + explainability + a hard cost floor

A learned market-price estimate combined with an inviolable sustainable-cost floor, plus a per-feature explanation of the recommendation.

[22] G. Ke, Q. Meng, T. Finley, et al., “LightGBM: A Highly Efficient Gradient Boosting Decision Tree,” in Proc. Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 3146–3154.

Applied in Kārigar for: The F4 price regressor (and the F5 quantile models).

[23] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in Proc. NeurIPS, 2017, pp. 4765–4774.

Applied in Kārigar for: SHAP values for the ‘Why this price?’ explanation.

[24] S. M. Lundberg, G. Erion, H. Chen, et al., “From Local Explanations to Global Understanding with Explainable AI for Trees,” Nature Machine Intelligence, vol. 2, pp. 56–67, 2020.

Applied in Kārigar for: Exact, fast tree-SHAP used on the trained LightGBM booster.

[25] A. V. den Boer, “Dynamic Pricing and Learning: Historical Origins, Current Research, and New Directions,” Surveys in Operations Research and Management Science, vol. 20, no. 1, pp. 1–18, 2015.

Applied in Kārigar for: Cost-plus vs. market-driven pricing trade-offs behind the blended recommendation.

[26] C. Molnar, Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, 2nd ed. Munich, 2022. [Online].

Applied in Kārigar for: Reference for presenting SHAP honestly to non-technical users.

E. F5 — Demand intelligence: probabilistic / quantile forecasting with uncertainty

Forecasts expressed as ranges, not single numbers, over clearly-labelled simulated history.

- [27] R. Koenker and G. Bassett, "Regression Quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.
Applied in Kārigar for: Quantile-regression basis for the P10 / P50 / P90 demand bands.
- [28] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, "DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks," *Int. Journal of Forecasting*, vol. 36, no. 3, pp. 1181–1191, 2020.
Applied in Kārigar for: Probabilistic-forecasting framing and evaluation methodology.
- [29] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting," *Int. Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
Applied in Kārigar for: Interpretable multi-horizon forecasting — documented upgrade path.
- [30] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M5 Accuracy Competition: Results, Findings, and Conclusions," *Int. Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.
Applied in Kārigar for: Retail-demand benchmark; why gradient boosting is a strong forecasting baseline.
- [31] Q. Wen, T. Zhou, C. Zhang, et al., "Transformers in Time Series: A Survey," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2023, pp. 6778–6786.
Applied in Kārigar for: Survey of modern forecasting models for the F5 roadmap.

F. F6 — Multi-artisan pooling: constraint optimisation + cooperative-game payment split

An exact MILP allocation under craft / capacity / logistics constraints, with a Shapley-value payment split (exact for small clusters, Monte-Carlo for large N).

- [32] L. S. Shapley, "A Value for n-Person Games," in *Contributions to the Theory of Games II (Annals of Mathematics Studies 28)*, Princeton Univ. Press, 1953, pp. 307–317.
Applied in Kārigar for: The fair payment-distribution rule for the pooled bulk order.
- [33] L. Perron, F. Didier, and S. Gay, "The CP-SAT-LP Solver," in *Proc. 29th Int. Conf. Principles and Practice of Constraint Programming (CP)*, 2023, pp. 3:1–3:2.
Applied in Kārigar for: The CP-SAT solver that runs the F6 allocation MILP.
- [34] J. Castro, D. Gómez, and J. Tejada, "Polynomial Calculation of the Shapley Value Based on Sampling," *Computers & Operations Research*, vol. 36, no. 5, pp. 1726–1730, 2009.
Applied in Kārigar for: Monte-Carlo Shapley estimator used when a cluster exceeds the exact-computation size.
- [35] R. Mitchell, J. Cooper, E. Frank, and G. Holmes, "Sampling Permutations for Shapley Value Estimation," *Journal of Machine Learning Research*, vol. 23, no. 43, pp. 1–46, 2022.
Applied in Kārigar for: Permutation-sampling design and variance behaviour for the estimator.
- [36] A. Aouadni, S. Aouadni, and A. Rebaï, "A Systematic Review on Supplier Selection and Order Allocation Problems," *Journal of Industrial Engineering International*, vol. 15, pp. 267–289, 2019.
Applied in Kārigar for: Formulation of multi-supplier order allocation as a constrained optimisation.
- [37] M. Guajardo and M. Rönnqvist, "A Review on Cost Allocation Methods in Collaborative Transportation," *Int. Transactions in Operational Research*, vol. 23, no. 3, pp. 371–392, 2016.
Applied in Kārigar for: Cooperative-game cost / benefit allocation for artisans sharing one order.

G. F7 — Fair market discovery: relevance retrieval + exposure-fairness re-ranking

TF-IDF / BM25 relevance combined with a bounded exposure boost for new and under-served sellers, without sacrificing relevance.

- [38] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
Applied in Kārigar for: Relevance-scoring basis for the discovery ranker.
- [39] A. Singh and T. Joachims, "Fairness of Exposure in Rankings," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, 2018, pp. 2219–2228.
Applied in Kārigar for: The exposure-fairness objective at the core of the F7 re-ranker.
- [40] A. J. Biega, K. P. Gummadi, and G. Weikum, "Equity of Attention: Amortizing Individual Fairness in Rankings," in *Proc. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2018, pp. 405–414.

Applied in Kārigar for: Amortised per-seller attention fairness across a stream of queries.

- [41] S. C. Geyik, S. Ambler, and K. Kenthapadi, “Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search,” in Proc. ACM SIGKDD, 2019, pp. 2221–2231.

Applied in Kārigar for: Production pattern for re-ranking under fairness constraints.

- [42] M. Zehlike, K. Yang, and J. Stoyanovich, “Fairness in Ranking, Part I: Score-Based Ranking; Part II: Learning-to-Rank and Recommender Systems,” ACM Computing Surveys, vol. 55, no. 6, pp. 1–41 / 1–42, 2022.

Applied in Kārigar for: Survey used to choose the fairness metric and its reporting.

- [43] G. K. Patro, A. Biswas, N. Ganguly, K. P. Gummadi, and A. Chakraborty, “FairRec: Two-Sided Fairness for Personalized Recommendations in Two-Sided Platforms,” in Proc. The Web Conf. (WWW), 2020, pp. 1194–1204.

Applied in Kārigar for: Two-sided (buyer and seller) fairness framing for the marketplace.

- [44] R. Mehrotra, J. McInerney, H. Bouchard, M. Lalmas, and F. Diaz, “Towards a Fair Marketplace: Counterfactual Evaluation of the Trade-off between Relevance, Fairness & Satisfaction in Recommendation Systems,” in Proc. ACM CIKM, 2018, pp. 2243–2251.

Applied in Kārigar for: Counterfactual evaluation of the exact relevance-vs-fairness trade-off F7 demonstrates.

H. F8 — Trust and verification: reputation, fact-checking, human-in-the-loop review

An evidence-backed verification state machine in which an AI risk flag is advisory only and a human reviewer makes the final decision.

- [45] A. Jøsang, R. Ismail, and C. Boyd, “A Survey of Trust and Reputation Systems for Online Service Provision,” Decision Support Systems, vol. 43, no. 2, pp. 618–644, 2007.

Applied in Kārigar for: Reliability-score and reputation design for artisans and buyers.

- [46] Z. Guo, M. Schlichtkrull, and A. Vlachos, “A Survey on Automated Fact-Checking,” Transactions of the Association for Computational Linguistics, vol. 10, pp. 178–206, 2022.

Applied in Kārigar for: The claim → evidence → verdict pipeline mirrored in the complaint workflow.

- [47] M. Steyvers and A. Kumar, “Three Challenges for AI-Assisted Decision-Making,” Perspectives on Psychological Science, vol. 19, no. 5, pp. 722–734, 2024.

Applied in Kārigar for: Why the AI flag stays advisory and the human reviewer decides.

- [48] M. Mitchell, S. Wu, A. Zaldivar, et al., “Model Cards for Model Reporting,” in Proc. ACM Conf. Fairness, Accountability, and Transparency (FAT*), 2019, pp. 220–229.

Applied in Kārigar for: Honest capability / limitation disclosure for every AI feature in the app.

- [49] T. Gebru, J. Morgenstern, B. Vecchione, et al., “Datasheets for Datasets,” Communications of the ACM, vol. 64, no. 12, pp. 86–92, 2021.

Applied in Kārigar for: Labelling of synthetic / demo training and history data.

I. Cross-cutting: digital public infrastructure, responsible AI, ICT for development

Context for a marketplace aimed at marginalised artisans and low-connectivity settings.

- [50] Open Network for Digital Commerce (ONDC), “ONDC Strategy Paper,” New Delhi, 2022. [Online].

Applied in Kārigar for: Interoperable-commerce context for the market-linkage design.

- [51] N. Sambasivan, S. Kapania, H. Highfill, D. Akrong, P. Paritosh, and L. M. Aroyo, “‘Everyone Wants to Do the Model Work, Not the Data Work’: Data Cascades in High-Stakes AI,” in Proc. ACM CHI Conf. Human Factors in Computing Systems, 2021, pp. 1–15.

Applied in Kārigar for: Why data quality and honest labelling are treated as first-class in the prototype.

- [52] B. Li, P. Qi, B. Liu, et al., “Trustworthy AI: From Principles to Practices,” ACM Computing Surveys, vol. 55, no. 9, pp. 1–46, 2023.

Applied in Kārigar for: Framework for the ‘what is REAL vs SIMULATED’ disclosures.